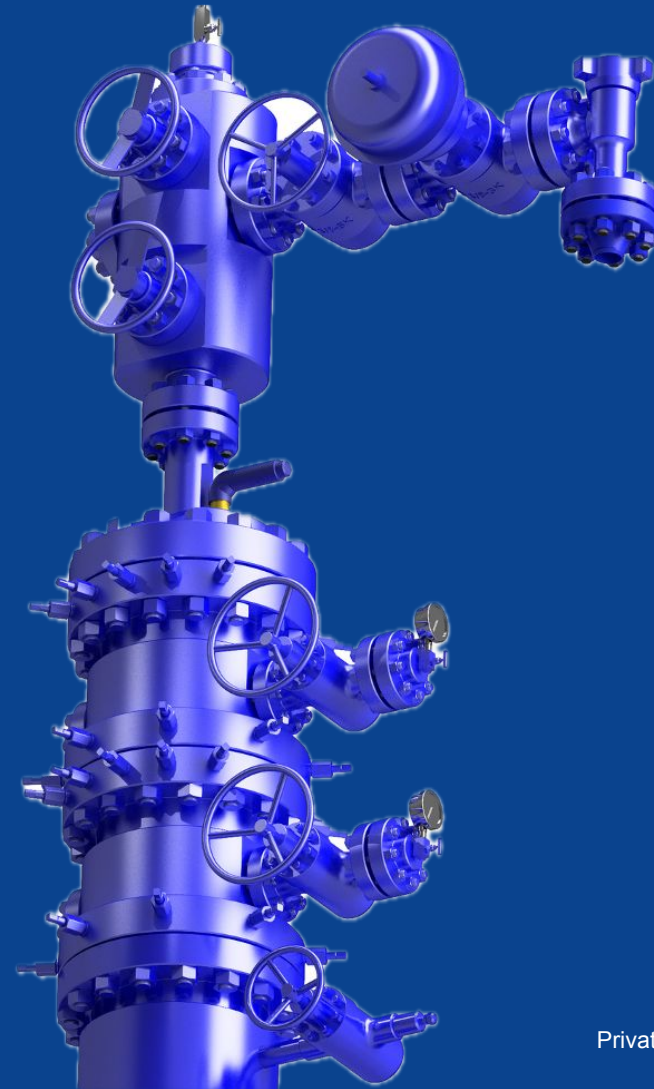


Wellhead & Xmas Tree Selection

10 July 2026





Trainer Introduction



20+ years work experience:

- ▶ Benvors, Wellhead Manufacturer
- ▶ Asia Serv, Wellhead Integrity
- ▶ TechnipFMC, Wellhead Manufacturer



References

- API 6A Specification for Wellhead and Tree Equipment
- <https://benvors.com/>
- <https://www.technipfmc.com/>
- <https://womgroup.com/>
- <https://sentrywellhead.com/>

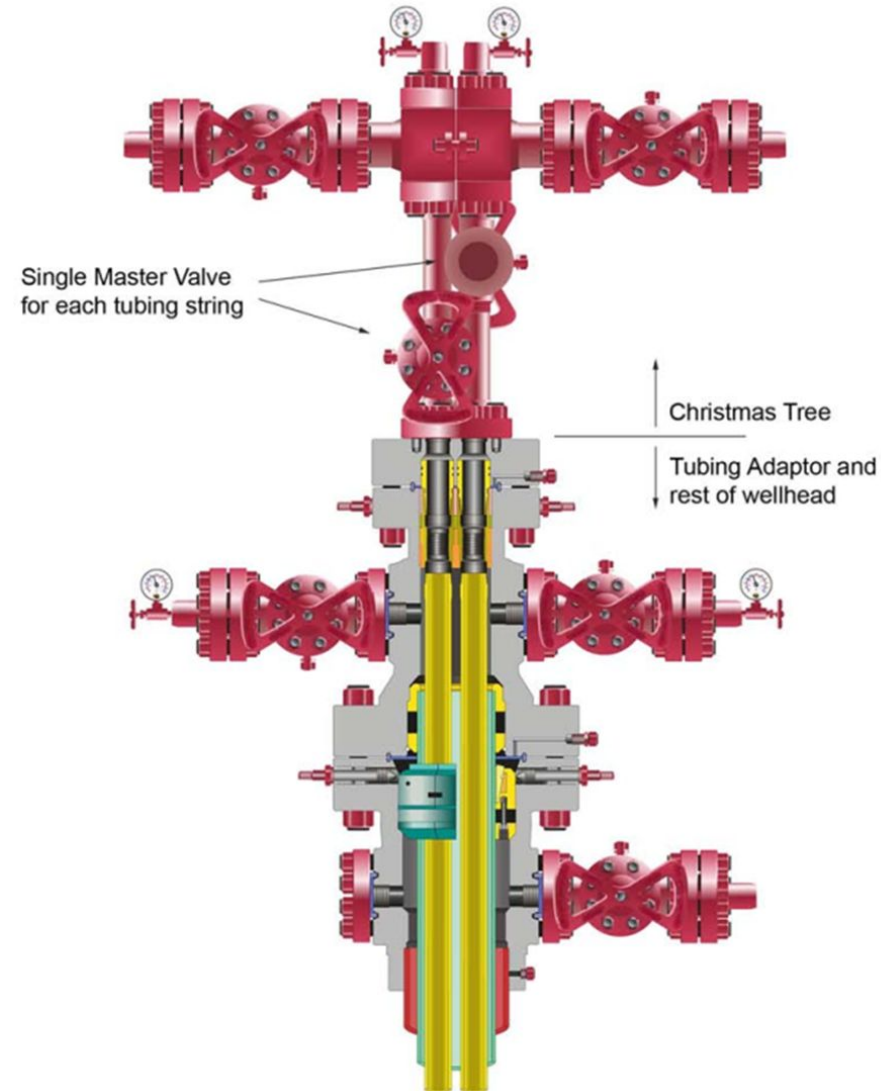


Outlines

1. Wellhead & Xmas Tree Overview
2. Oil & Gas Wellhead Component
3. Wellhead Accessories & Special Equipment
4. Geothermal Wellhead Component
5. Wellhead Sizing & Rating
6. Testing, Codes & Standards
7. Wellhead Selection
8. Case Study
9. Q & A



Wellhead and Xmas Tree Overview





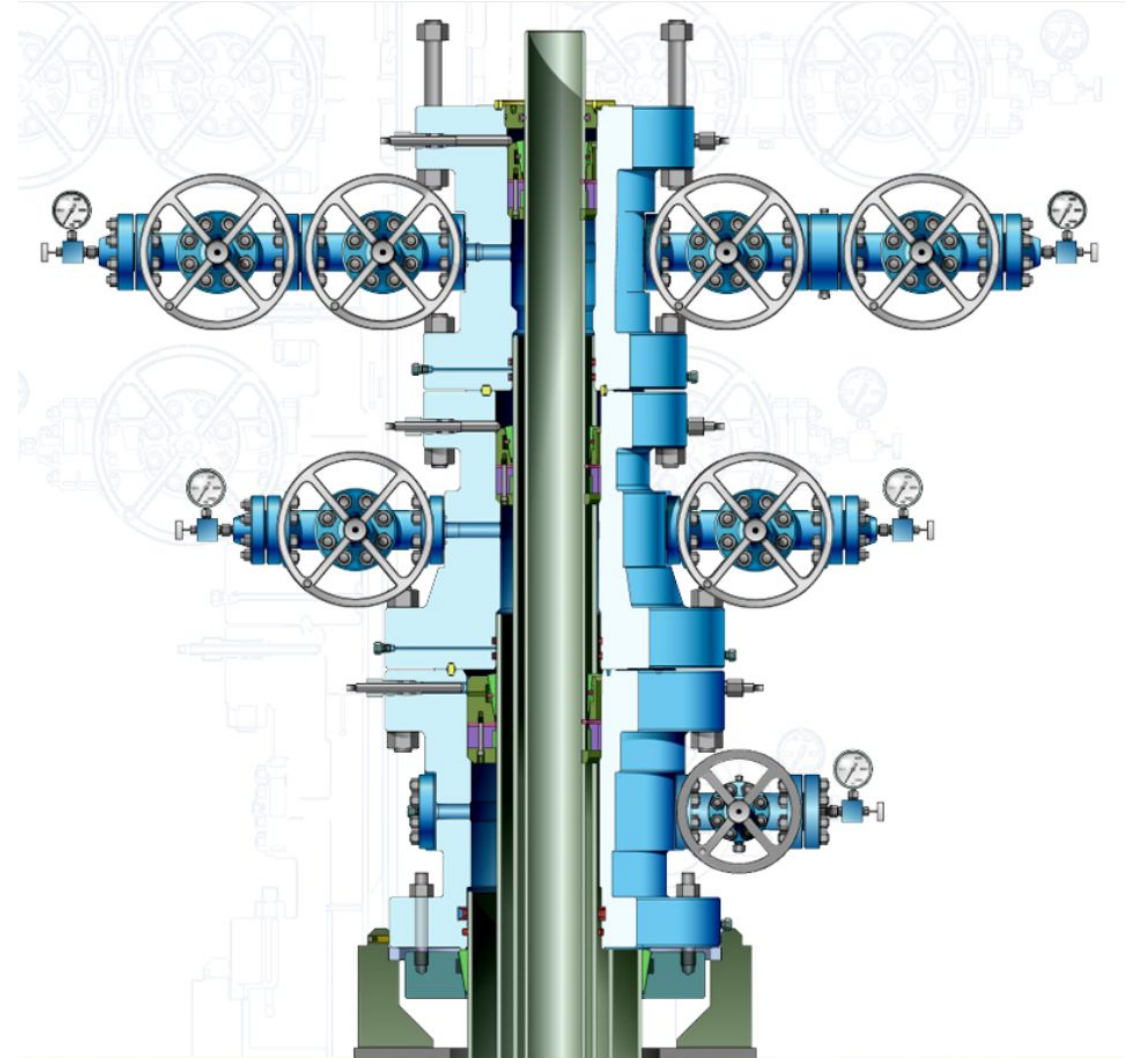
Wellhead

What is a Wellhead?

A pressure-containing equipment installed at surface to suspend casing strings, seal annular pressure, and provide pressure control throughout the life of the well.

Main Functions

- Structural support
- Pressure containment
- Casing suspension
- Well integrity
- Interface to Christmas Tree
- Access for drilling and intervention

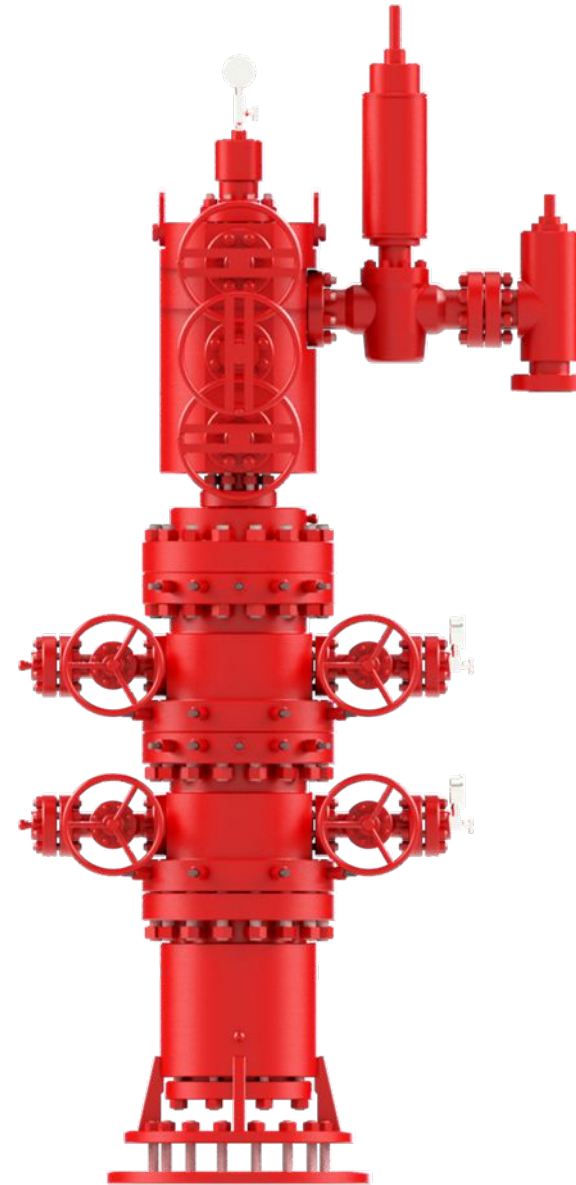




Wellhead Component

Major Components:

- Casing Head
- Casing Spool
- Casing Hanger
- Tubing Head
- Tubing Hanger
- Tubing Head Adapter

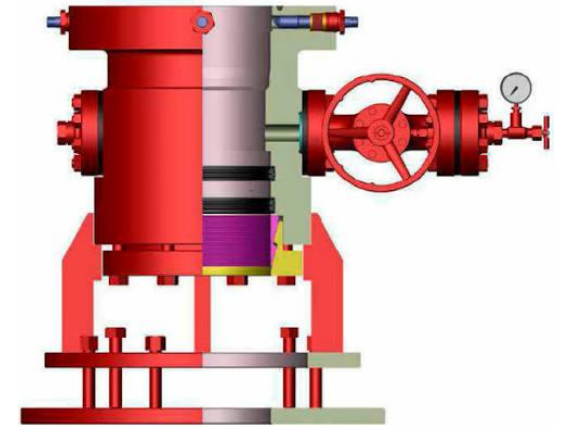
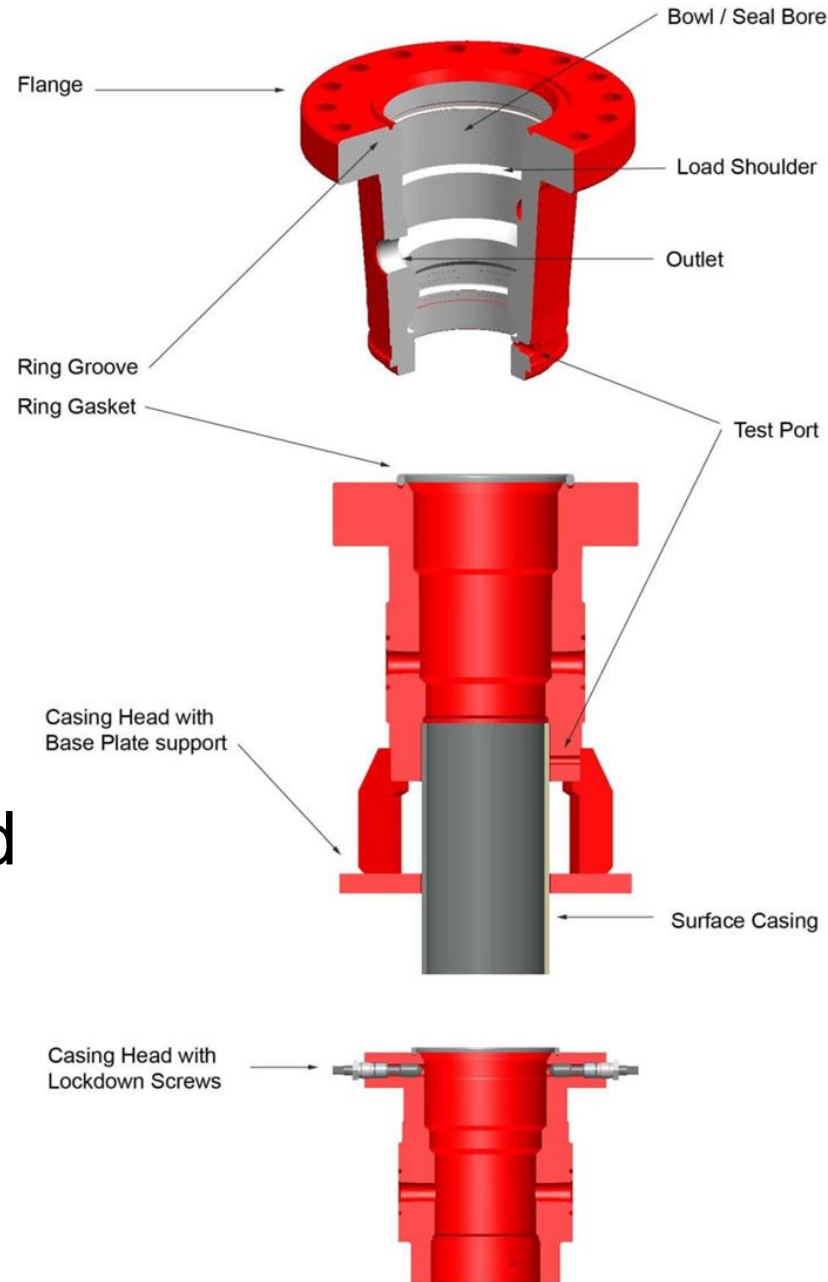




Casing Head

Function

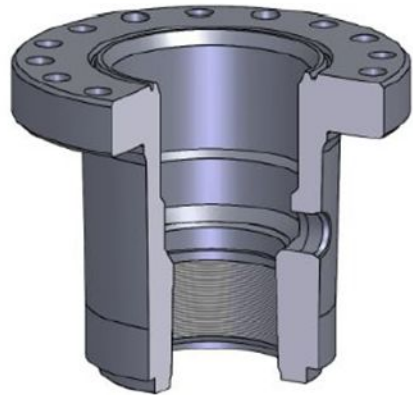
- Suspend casing
- Seal annulus
- Provide outlet
- Pressure containment
- Connect upper wellhead component



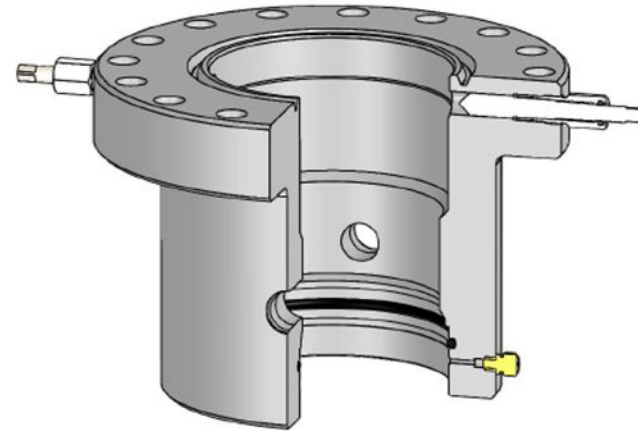


Casing Head

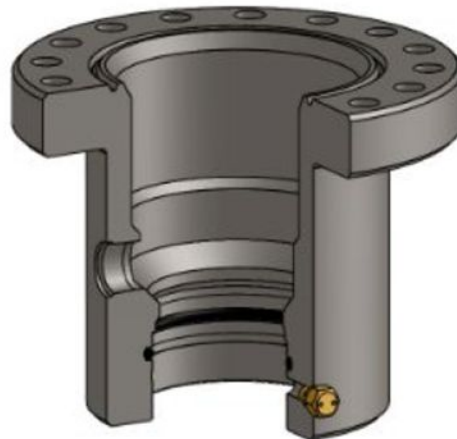
C-22 Casing Head Configurations



C-22 W/Threaded Bottom & Outlets



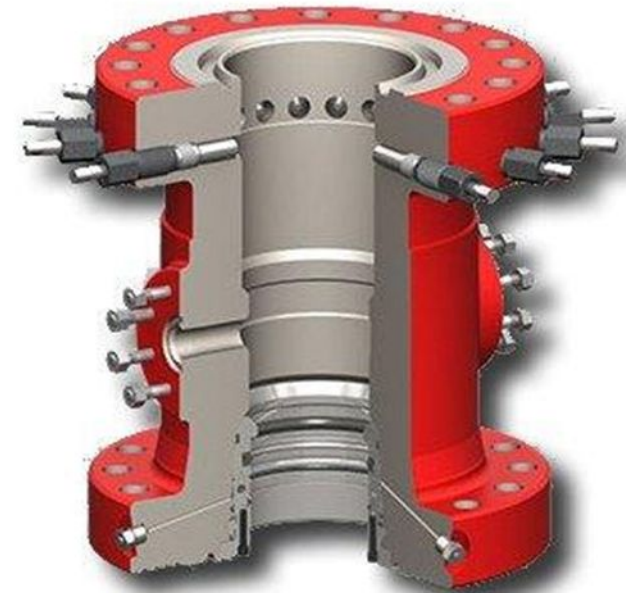
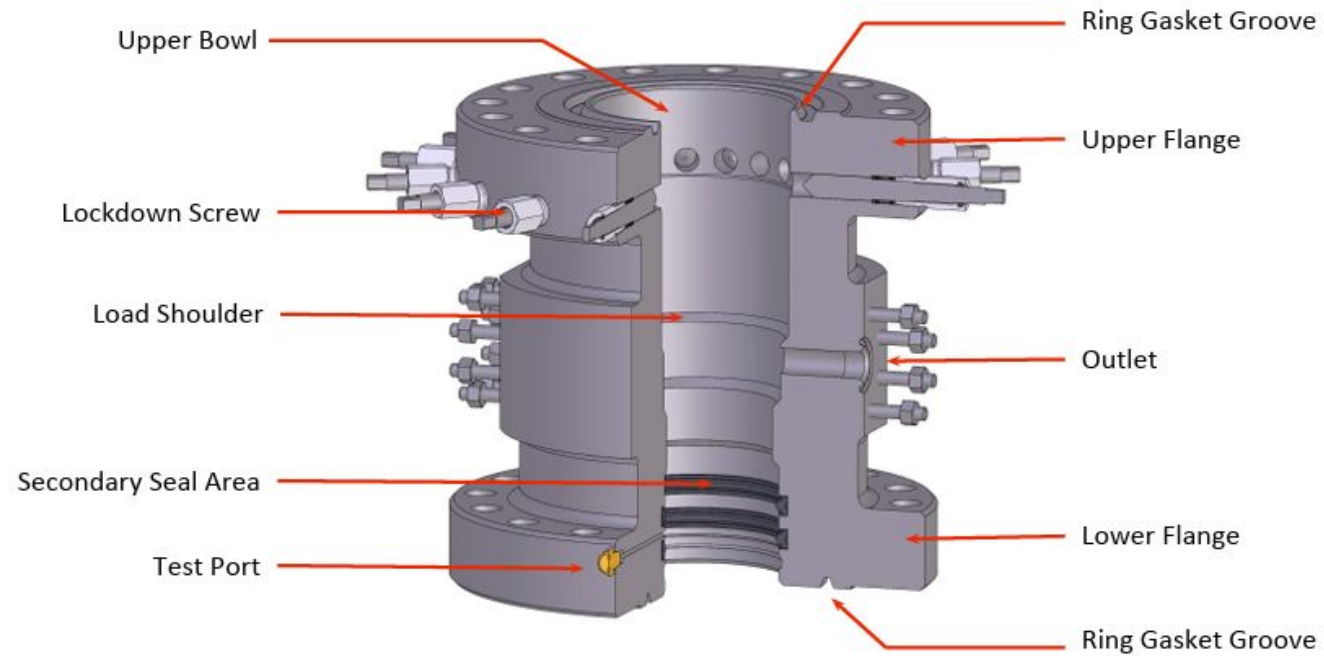
C-22-BP-ET W/SOW Bottom & Threaded Outlets



C-22 W/SOW Bottom & Threaded Outlets



Casing Spool



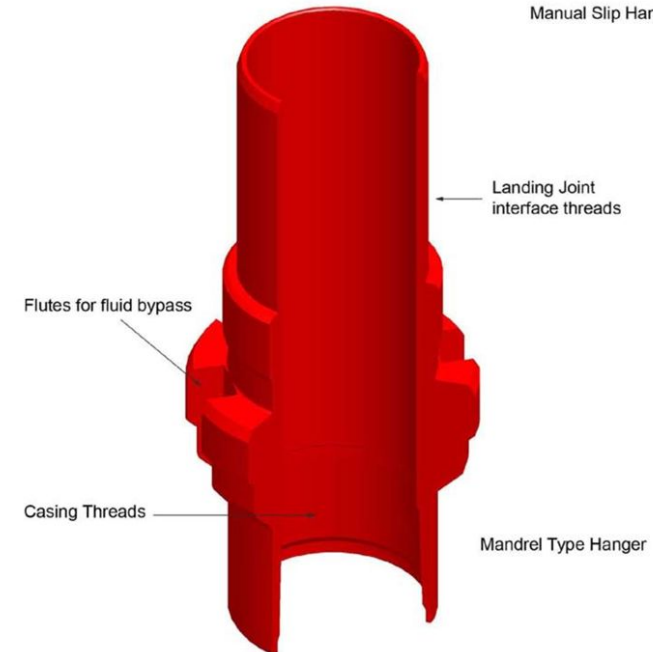
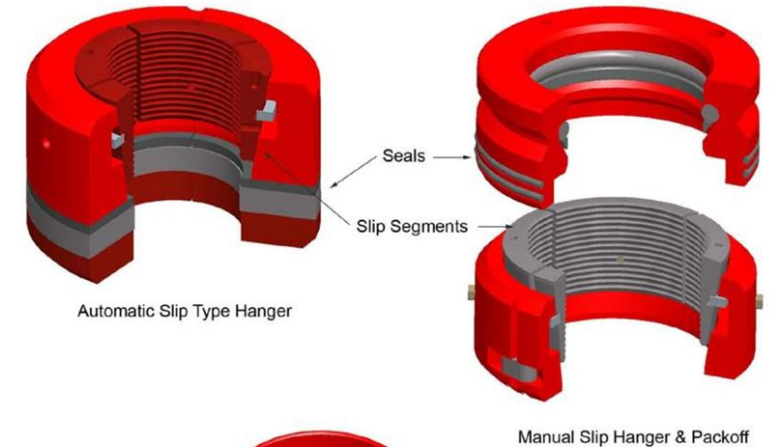
Casing Hanger

Function

- To suspend the load of the casing string from the casing head or spool.
- To center the casing in the head.
- To provide a primary seal against the inside of the casing head and isolate the casing annulus pressure from upper wellhead components.

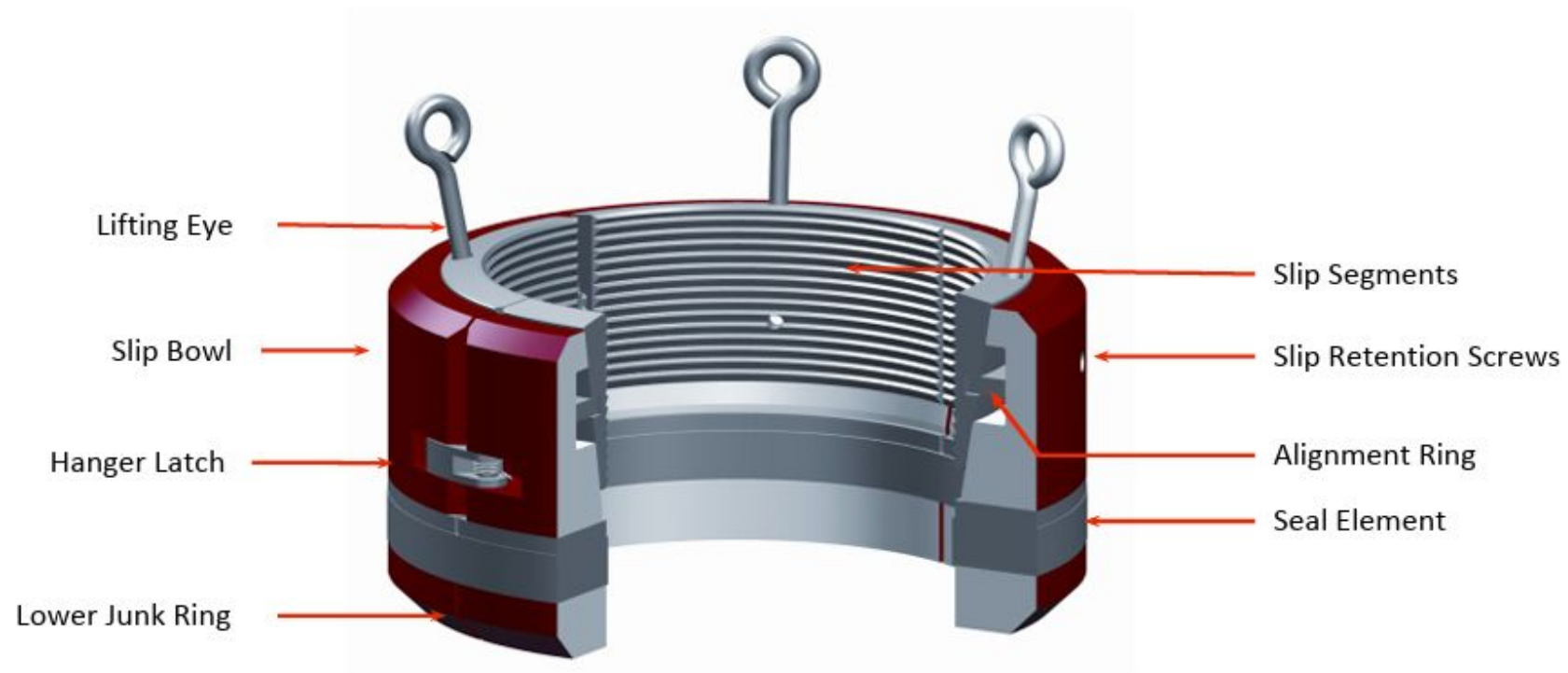
Types

- Slip Type
- Mandrel Type





Casing Hanger Components

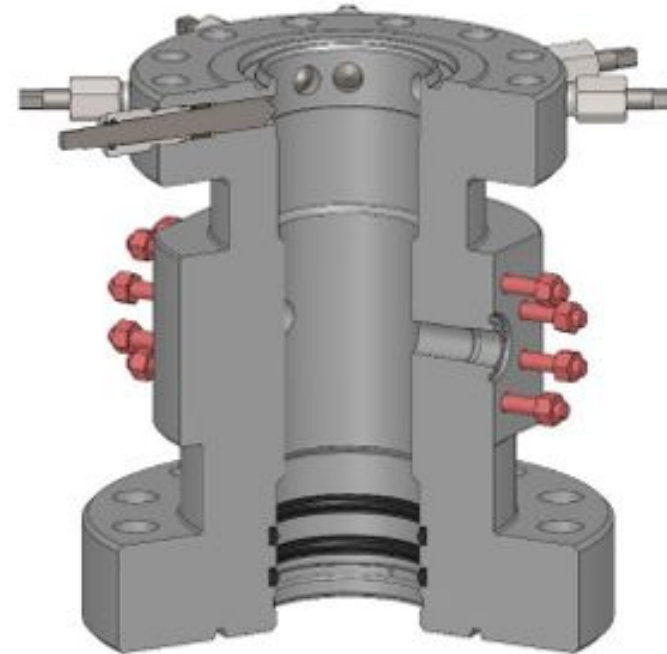




Tubing Head

Function

- Support tubing
- Seal tubing annulus
- Provide production outlets
- Provide access to tubing hanger



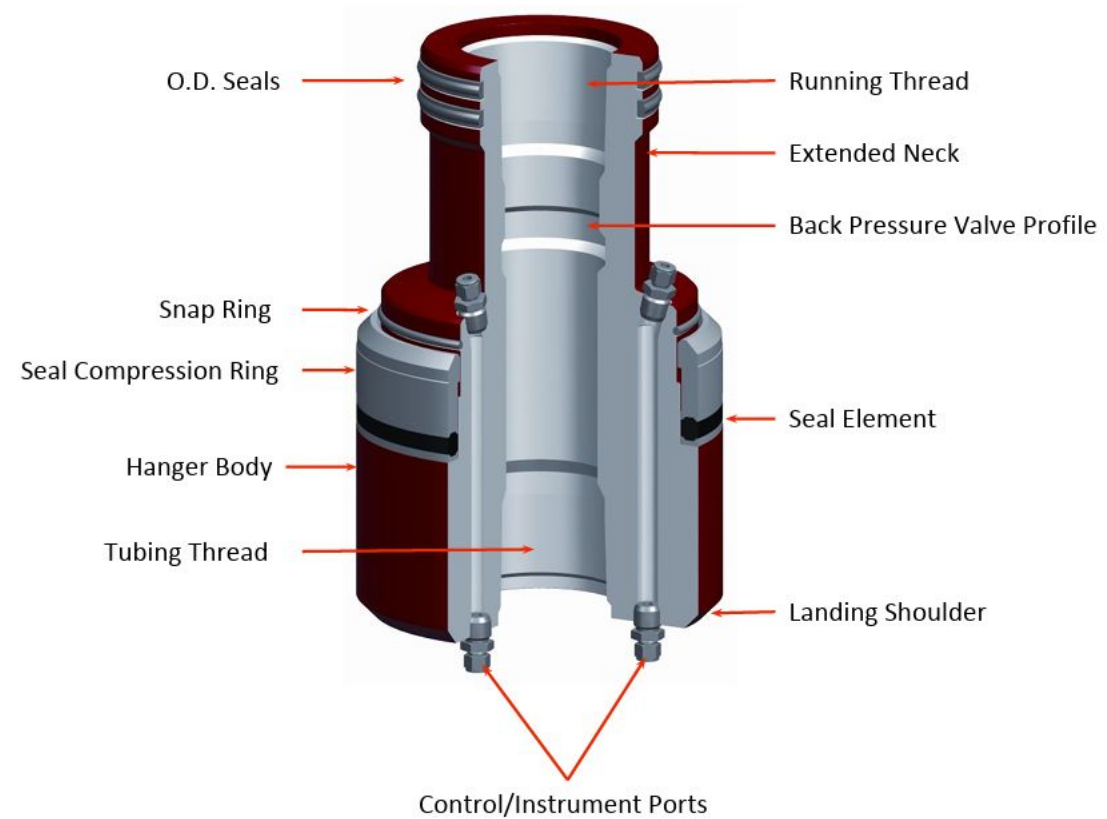


Tubing Hanger

Types

- Mandrel
- Wrap Around
- Dual Completion
- ESP Completion

Tubing Hanger Components





Tubing Hanger



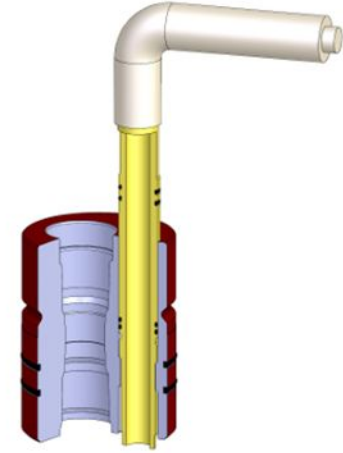
Mandrel
Type



Wraparound
Type



Dual Completion

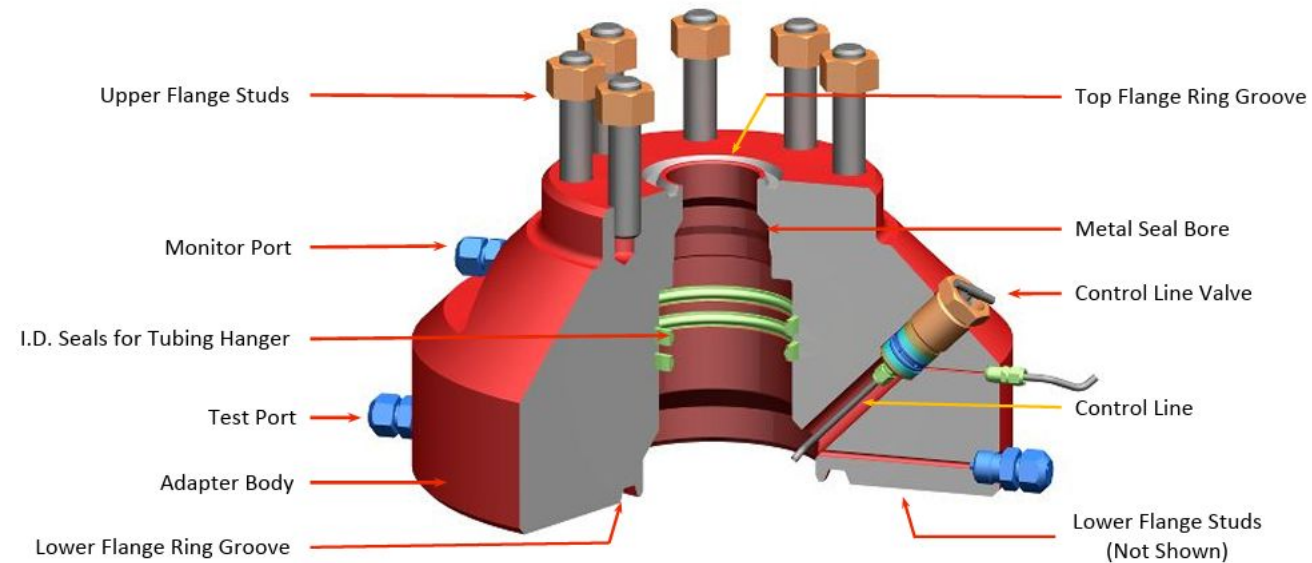


ESP Completion



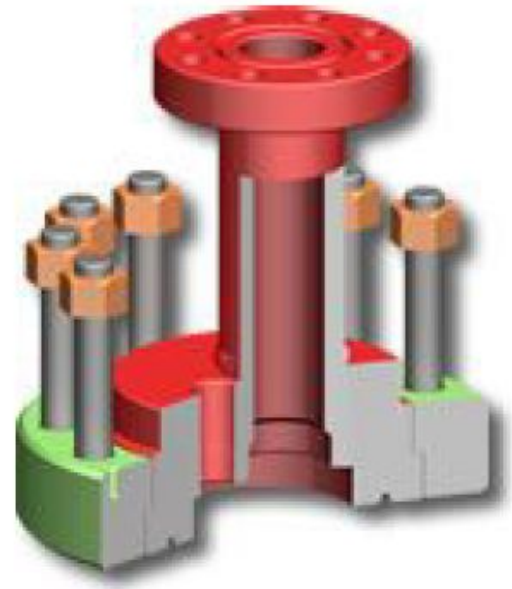
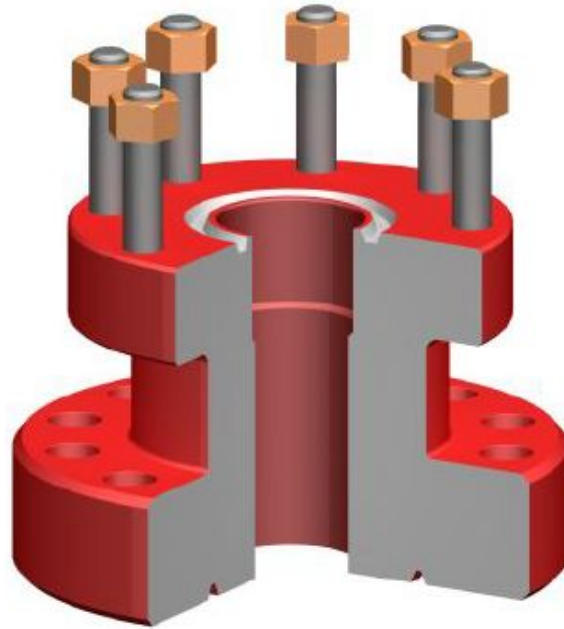
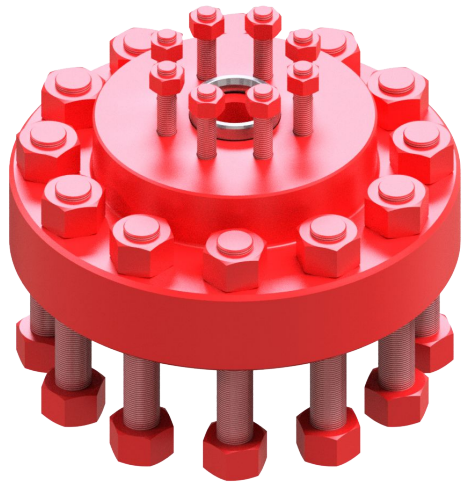
Tubing Head Adapter

Tubing Head Adapter Components

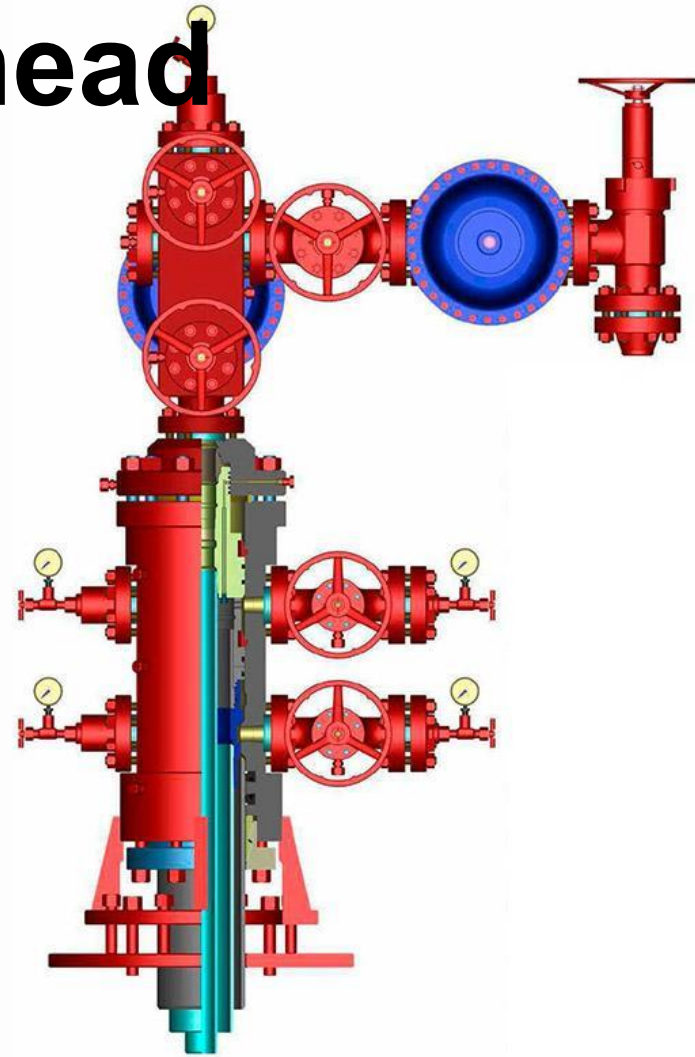
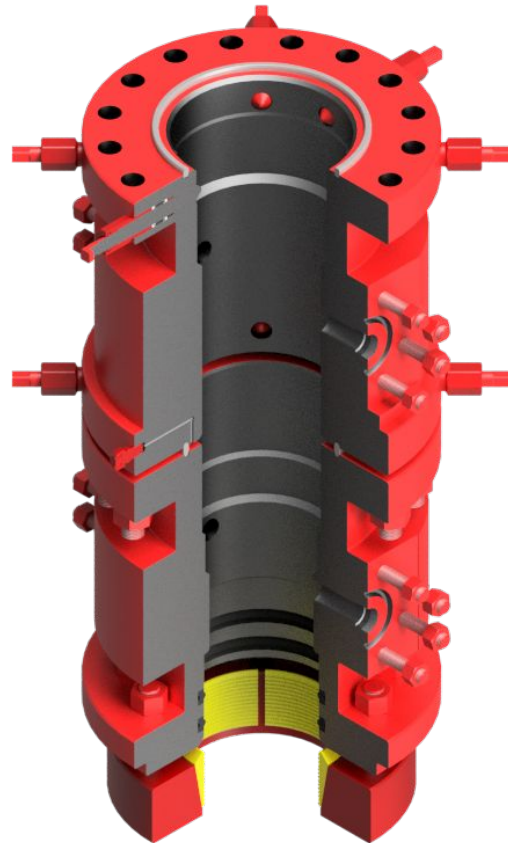




Tubing Head Adapter



Unihead (Compact) Wellhead

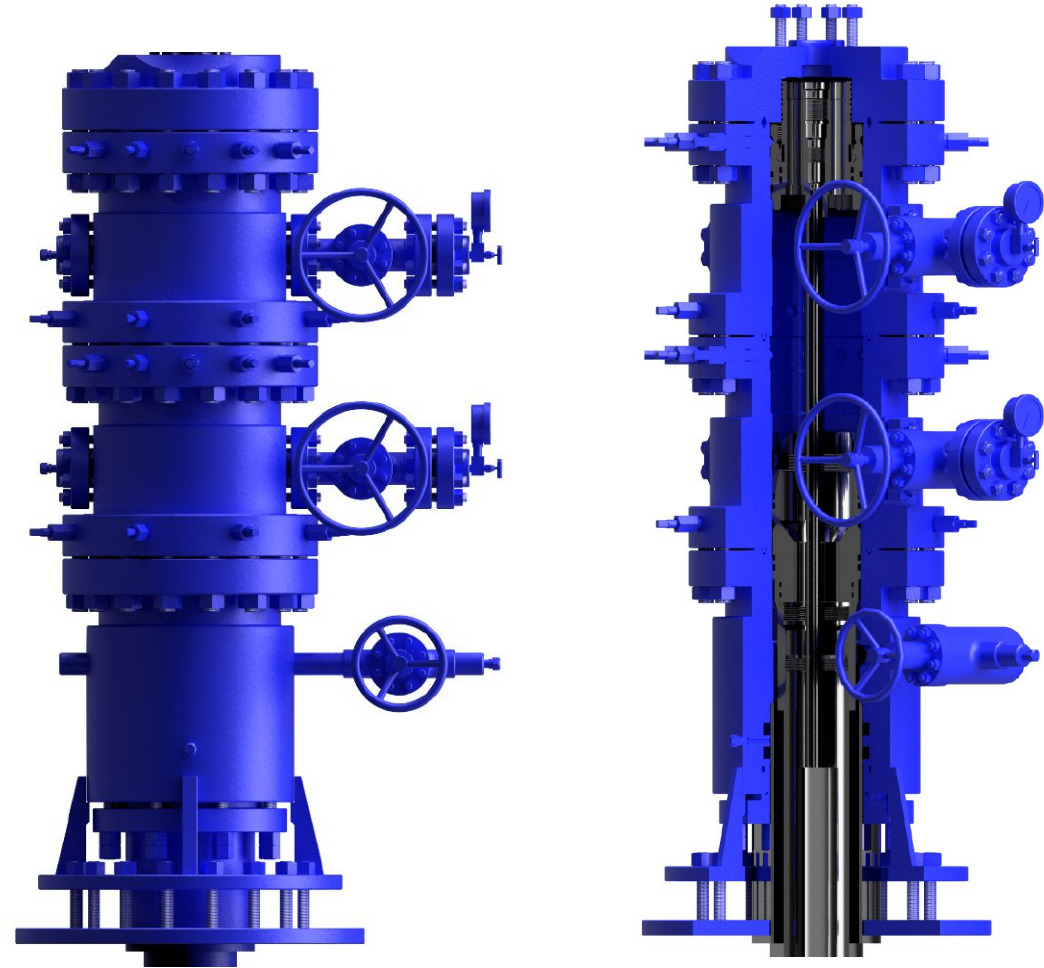




Unihead/Compact/Integrated Wellhead

The Unihead Wellhead System significantly enhances user experience by delivering a unified solution that ensures improved safety, accelerated operational processes, and cost-effectiveness. This streamlined approach provides users with a safer, faster, and more economical method for well drilling.

- Saving rig time could be 2 - 3 days
- Multiplier effect on drilling economics: reducing expenditures on standby equipment and personnel when nipple up/nipple down BOP and WOC
- By eliminating steps on reducing nipple-up and nipple-down BOP, it eliminate multiple risk, reducing manhours on the cellar area, making it safer





Time Saving

Conventional Wellhead	Time (Hours)	Integrated Wellhead	Time (Hours)
Run and weld casing head nipple up BOP	12	Run Integrated Casing Head	12
Run and cement 9 5/8" casing, nipple down BOP, set slips, wait on cement and install next head and BOP	50	Run 9 5/8" casing and packoff (no BOP removal or installation of the next head)	14
Run 7" casing, nipple down BOP, set slips, wait on cement and install next head and BOP	50	Run 7" casing and packoff (no BOP removal or installation of the next head)	10
Install x-mas tree	4	Install x-mas tree	4
Total Time	116	Total Time	40

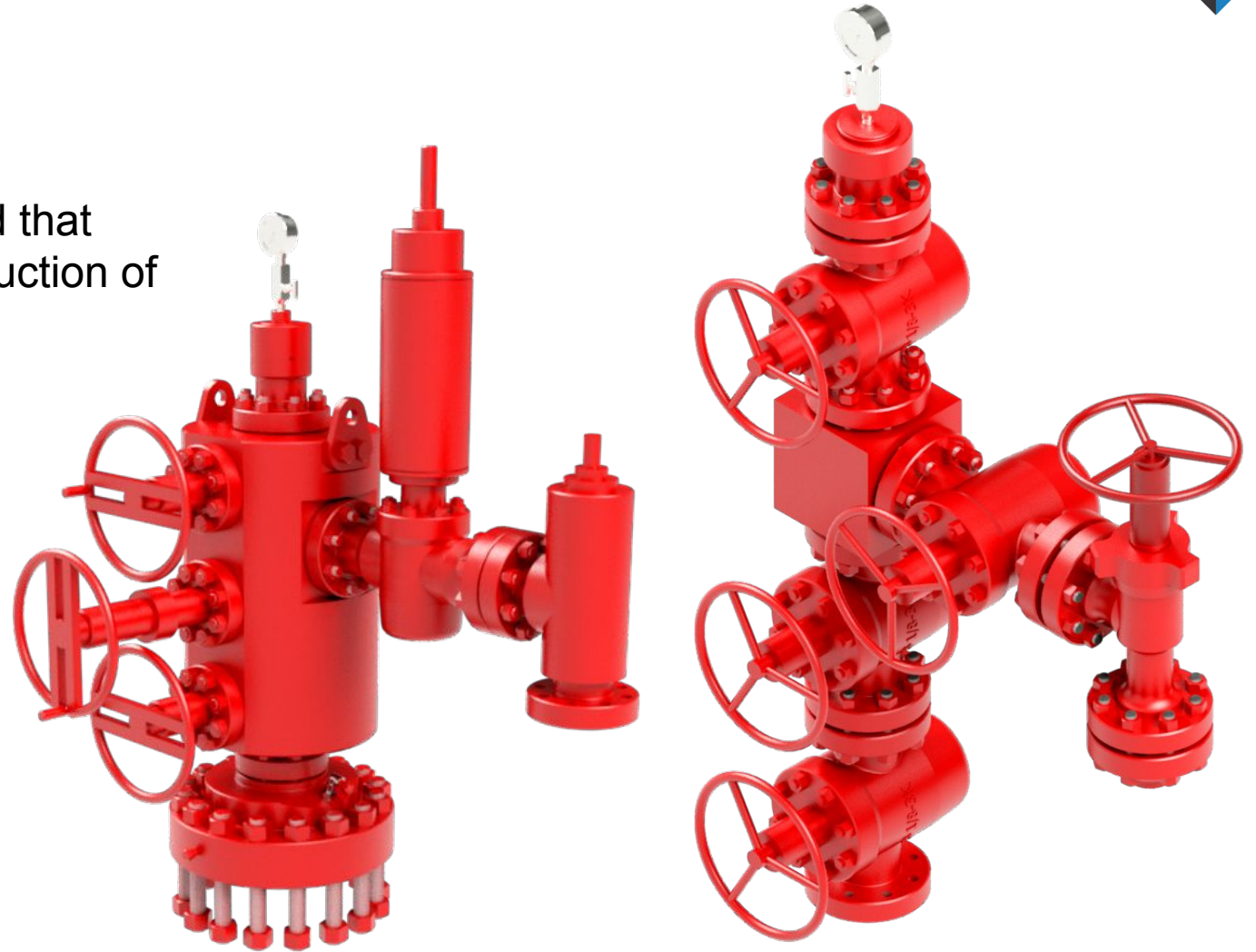


Xmas Tree

A Christmas Tree is the pressure control equipment installed on top of the wellhead that regulates, isolates, and monitors the production of oil, gas, or injected fluids from a well.

Components

- Master Valve
- Wing Valve
- Swab Valve
- Tree Cap
- Choke



Wellhead Accessories – Wear bushing



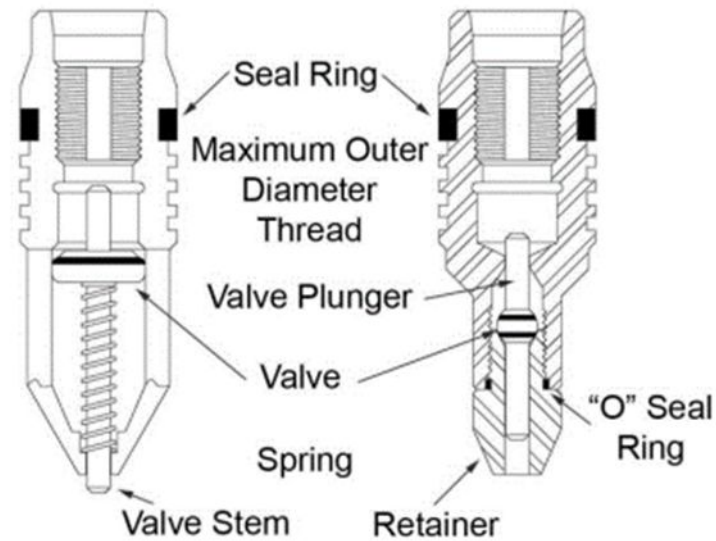
- While drilling the well, it is required that the seal bores in each of the intermediate casing spools and tubing spools be protected.
- A series of wear bushings are supplied to protect the seal areas discussed during the drilling operation.
- The wear bushings are run on a drill pipe tool with J-lugs located on the OD that interface with J-slots located in the top ID section of the wear bushing.



Wellhead Accessories – BPV & TWCV



- Tubing hangers may come with a back pressure thread profile that enables the operator to install an isolation plug into the tubing hanger



Threaded Back Pressure Valves



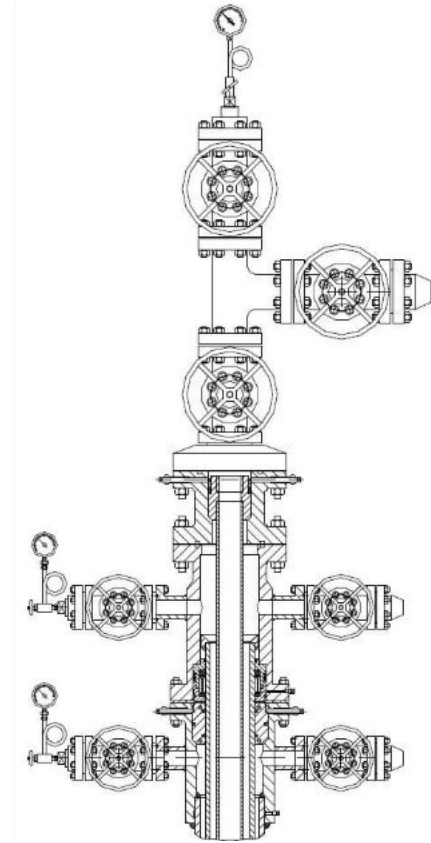
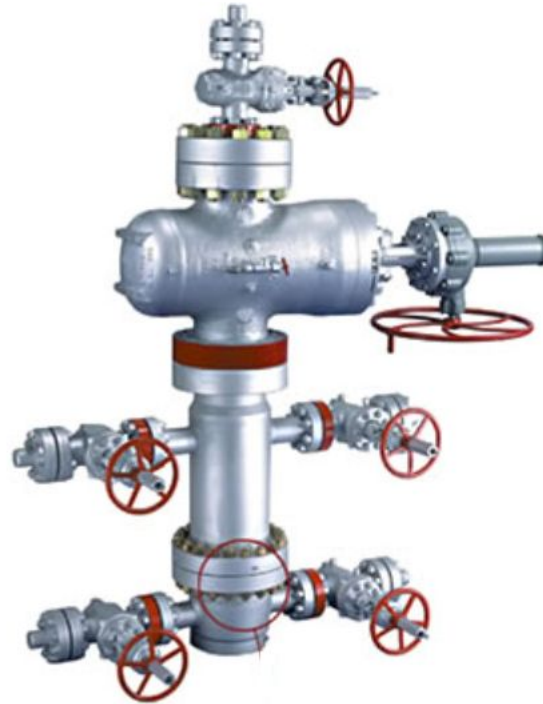
IS-A
Back Pressure Valve



Geothermal Wellhead

Why Geothermal Wellhead is Different

- Steam
- High Temperature
- Thermal Expansion
- Scaling
- Corrosion
- Cycling

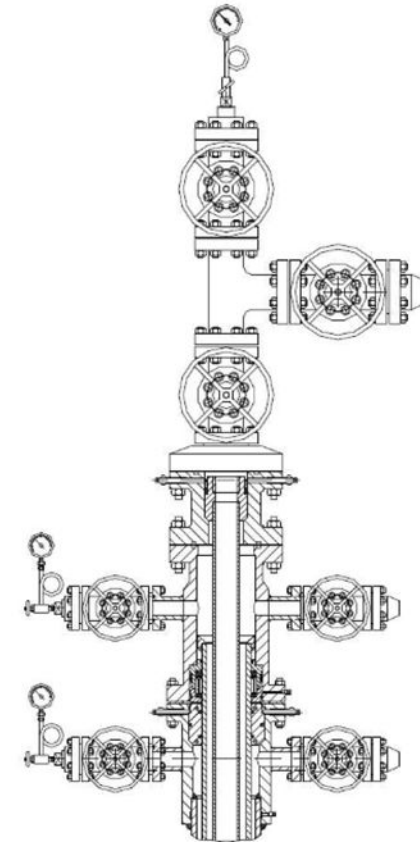
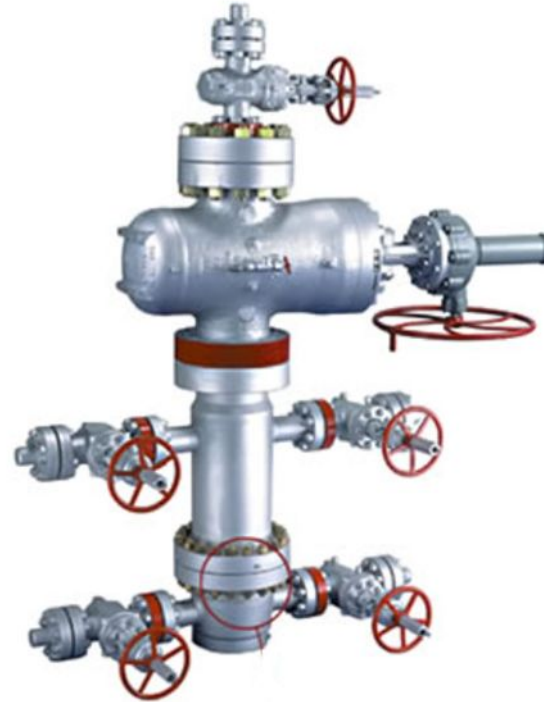




Geothermal Wellhead

Typical Geothermal Wellhead

- Master Valve
- Expansion Spool
- Steam Cross
- Production Valve



Geothermal vs Oil & Gas Wellhead Comparison



Aspect	Oil & Gas Wellhead	Geothermal Wellhead	Design Impact
Steam / Production Fluid	Oil, gas, or multiphase hydrocarbons	Dry steam, wet steam, and hot brine (two-phase flow)	Full-bore flow path, erosion-resistant valves and fittings
High Temperature	Typically up to ~120°C in conventional applications	Typically 180–350°C (or higher)	High-temperature materials, metal-to-metal sealing, temperature-resistant components
Thermal Expansion	Relatively limited thermal growth	Significant thermal expansion due to continuous high temperatures	Expansion spool, flexible supports, thermal stress management
Scaling	Generally minimal	Silica, calcium carbonate, and other mineral deposits	Easy maintenance access, scale-resistant flow paths, periodic cleaning
Corrosion	Depends on produced fluids	High-temperature corrosive environment (H ₂ S, CO ₂ , chlorides)	Corrosion-resistant materials, cladding, special valve trims
Thermal Cycling	Less frequent temperature fluctuations	Repeated start-up/shut-down heating and cooling cycles	Fatigue-resistant design, robust bolting, reliable sealing system



Testing, Codes & Standards

Factory Testing

- Hydrostatic Test
- Gas Test
- Shell Test
- Seat Test
- Function Test

Quality Documentation

- Inspection & Test Plan (ITP)
- Inspection Record
- Pressure Test Chart
- Dimensional Report
- Material Certificate
- Certificate of Conformance (COC)

FOOTNOTE

^a For flange sizes 26³/₄ in. and 30 in., see Table D.7.



Pressure Testing

Table 33—Hydrostatic Shell Test Pressure

Working Pressure Rating	Hydrostatic Shell Test Pressure						
	Nominal Size of Flange mm (in.)		Line Pipe and Tubing Threads	Casing Threads mm (in.)			
	346 (13 ⁵ / ₈) and smaller	425 (16 ³ / ₄) and larger		114.3 to 273.1 (4 ¹ / ₂ to 10 ³ / ₄)	298.5 to 339.7 (11 ³ / ₄ to 13 ³ / ₈)	406.5 to 508.0 (16 to 20)	
MPa (psi)	MPa (psi)	MPa (psi)	MPa (psi)	MPa (psi)	MPa (psi)	MPa (psi)	MPa (psi)
13.8 (2000)	27.6 (4000)	20.7 (3000)	27.6 (4000)	27.6 (4000)	27.6 (4000)	27.6 (4000)	15.5 (2250)
20.7 (3000)	41.4 (6000)	31.0 (4500)	41.4 (6000)	41.4 (6000)	41.4 (6000)	31.0 (4500)	— —
34.5 (5000)	51.7 (7500)	51.7 (7500)	51.7 (7500)	51.7 (7500)	51.7 (7500)	— —	— —
69.0 (10,000)	103.5 (15,000)	103.5 (15,000)	103.5 (15,000)	— —	— —	— —	— —
103.5 (15,000)	155.0 (22,500)	155.0 (22,500)	— —	— —	— —	— —	— —
138.0 (20,000)	207.0 (30,000)	— —	— —	— —	— —	— —	— —



Temperature Rating

4.3.2 Temperature Ratings

Equipment shall be designed to operate in one or more of the specified temperature classes with minimum and maximum temperatures as shown in Table 2, or to minimum and maximum temperature ratings as agreed between the purchaser and manufacturer.

Table 2—Temperature Ratings

Temperature Class	Temperature Range			
	°C		°F	
	min.	max.	min.	max.
K	−60	82	−75	180
L	−46	82	−50	180
N	−46	60	−50	140
P	−29	82	−20	180
S	−18	60	0	140
T	−18	82	0	180
U	−18	121	0	250
V	2	121	35	250
NOTE Minimum temperature is the lowest ambient temperature to which the equipment can be subjected. Maximum temperature is the highest temperature of the fluid that can directly contact the equipment.				

Design for temperature ratings above 121 °C (250 °F), e.g. classifications X and Y (see Table G.1), shall account for the effects of temperature on material strength (see Annex G for guidelines).



Temperature Rating

G.2 Elevated Temperature Ratings

NOTE The temperature ratings given in Table G.1 may be used for equipment for service temperatures in excess of those covered by Section 4.

Table G.1—Temperature Ratings

Classification	Temperature Range	
	°C	°F
X	-18 to 180	0 to 350
Y	-18 to 345	0 to 650



Material Class

Table 3—Material Requirements

Material Class		Body, Bonnet, End and Outlet Connectors	Mandrel Hangers, Valve Bore Sealing Mechanisms, Choke Trim, and Stems
AA	General service	Carbon or low-alloy steel, or stainless steel or CRA ^d	Carbon or low-alloy steel, or stainless steel or CRA ^d
BB	General service	Carbon or low-alloy steel, or stainless steel or CRA ^d	Stainless steel or CRA ^d
CC	General service	Stainless steel or CRA ^d	Stainless steel or CRA ^d
DD	Sour service ^a	Carbon or low-alloy steel or CRA ^{b,d}	Carbon or low-alloy steel, or stainless steel or CRA ^{b,d}
EE	Sour service ^a	Carbon or low-alloy steel or CRA ^{b,d}	Stainless steel or CRA ^{b,d}
FF	Sour service ^a	Stainless steel or CRA ^{b,d}	Stainless steel or CRA ^{b,d}
HH	Sour service ^a	CRA ^{b,c,d}	CRA ^{b,c,d}
FOOTNOTES			
^a As defined by NACE MR0175/ISO 15156.			
^b In accordance with NACE MR0175/ISO 15156 and 4.3.3.2.			
^c CRA required on retained fluid-wetted surfaces only; CRA cladding of low-alloy or stainless steel is permitted (see 7.5.1.2).			
^d CRA as defined in 3.1.22; NACE MR0175/ISO 15156 definition of CRA does not apply.			

Material Classes for Sour Service

For material classes DD, EE, FF, and HH, the manufacturer shall meet the requirements of NACE MR0175/ISO 15156 for material processing and material properties (e.g. hardness).



Material Class

Table B.1—Relative Corrosivity of Retained Fluids as Indicated by CO₂ Partial Pressure

Retained Fluids	Relative Corrosivity	Partial Pressure of CO ₂	
		MPa	psia
General service	Noncorrosive	< 0.05	< 7
General service	Slightly corrosive	0.05 to 0.21	7 to 30
General service	Moderately to highly corrosive	> 0.21	> 30
Sour service	Noncorrosive	< 0.05	< 7
Sour service	Slightly corrosive	0.05 to 0.21	7 to 30
Sour service	Moderately to highly corrosive	> 0.21	> 30



Material Class

Table F.2—Standard Test Fluids for Nonmetallic Seals

Material Class	Hydrocarbon Liquid Phase ^a	Gas Phase
AA/BB	b	5 % vol. fraction CO ₂ /95 % vol. fraction CH ₄
CC	b	80 % vol. fraction CO ₂ /20 % vol. fraction CH ₄
DD/EE	b	10 % vol. fraction H ₂ S/5 % vol. fraction CO ₂ /85 % vol. fraction CH ₄
FF/HH	b	10 % vol. fraction H ₂ S/80 % vol. fraction CO ₂ /10 % vol. fraction CH ₄
FOOTNOTES		
^a Water shall be added to the liquid phase.		
^b Hydrocarbon liquid phase selected at the manufacturer's discretion may include, but is not limited to, jet fuel, diesel, kerosene, etc.		



Product Specification Level (PSL)

PSL designations shall define different levels of technical quality requirements as identified in this specification. Products manufactured to the requirements of this specification shall satisfy the material, welding, quality, and testing requirements for a PSL (PSL 1, PSL 2, PSL 3, and PSL 4), when applicable.

PSLs shall be applied to products as designated in Table 4.



Product Specification Level (PSL)

Table 4—Applicability of Product Specification Levels

Equipment Category and Type (Reference Section)	Applicable PSLs	Equipment Category and Type (Reference Section)	Applicable PSLs
Plugs, Connectors, Gaskets		Valves and Chokes	
Flanges (blind, test) ^a (see 14.1)	1, 2, 3, 4	Valves (gate, plug, ball) (see 14.11)	1, 2, 3 ^e , 4
Ring gaskets ^b (see 10.4.5 and 14.2)	NA	Valves (prepared for/and actuated) (see 14.11)	1, 2, 3 ^e , 4
Threaded connectors ^a (see 14.3)	1, 2, 3, 4	Check valves (see 14.11)	1, 2, 3 ^e , 4
Tees and crosses (see 14.4)	1, 2, 3 ^e , 4	Back-pressure valves ^b (see 14.12)	NA
Bullplugs ^b (see 14.5)	NA	SSVs and USVs ^c (see 14.17)	2, 3 ^e , 4
Valve-removal plugs ^b (see 14.6)	NA	BSDVs ^d (see 14.17)	3 ^e , 4
Top connectors (see 14.7)	1, 2, 3 ^e , 4	Chokes (adjustable and positive) (see 14.15)	1, 2, 3 ^e , 4
Crossover connectors (see 14.8)	1, 2, 3 ^e , 4	Casing and Tubing Heads	
Other end connectors ^a (see 14.9)	1, 2, 3, 4	Housings (see 14.14)	1, 2, 3 ^e , 4
Spools (adapter, spacer) (see 14.10)	1, 2, 3 ^e , 4	Adapters (see 14.14)	1, 2, 3 ^e , 4
Weld-neck flanges ^a (see J.1)	1, 2, 3, 4	Other Equipment	
Segmented flanges ^a (see L.1)	1, 2, 3, 4	Actuators ^b (see 14.16)	NA
Nonintegral metal seals ^a (see 10.4.5)	NA	Tree assemblies ^b (see 14.18)	NA
Casing and Tubing Hangers		Packing mechanisms ^b (see 9.1)	NA
Slip-type ^a (see 14.13)	1, 2, 3, 4	Pressure boundary penetrations ^b (see 9.2)	NA
Mandrel-type ^a (see 14.13)	1, 2, 3, 4	Test and gauge ports ^b (see 9.3)	NA
FOOTNOTES ^a Gas testing is not required, so PSL 3G designation is not applicable. ^b There is only one level of requirements for these products, so PSLs are not applicable (NA). ^c PSL 1 is not applicable to SSVs and USVs. ^d PSL 1 and PSL 2 are not applicable to BSDVs. ^e For products eligible for gas testing, PSL 3G designation and marking may apply.			



Material Class	Rated Working Pressure (RWP)					
	13.8 MPa (2000 psi)	20.7 MPa (3000 psi)	34.5 MPa (5000 psi)	69.0 MPa (10,000 psi)	103.5 MPa (15,000 psi)	138.0 MPa (20,000 psi)
AA, BB, CC	PSL 1, PSL 2, PSL 3, PSL 4	PSL 1, PSL 2, PSL 3, PSL 4	PSL 1, PSL 2, PSL 3, PSL 4	PSL 2, PSL 3, PSL 4	PSL 2, PSL 3, PSL 4	PSL 3, PSL 4
DD, EE, FF	PSL 1, PSL 2, PSL 3, PSL 4	PSL 1, PSL 2, PSL 3, PSL 4	PSL 1, PSL 2, PSL 3, PSL 4	PSL 2, PSL 3, PSL 4	PSL 3, PSL 4	PSL 3, PSL 4
HH, ZZ	PSL 3, PSL 4	PSL 3, PSL 4	PSL 3, PSL 4	PSL 3, PSL 4	PSL 3, PSL 4	PSL 4

NOTE 1 For RWP/Material Class combinations that may have more than one applicable PSL, only one PSL is marked on products that satisfy the material, welding, quality, and testing requirements for that specific PSL (see 4.3.4.1).

NOTE 2 For products eligible for gas testing, PSL 3G may be applied wherever PSL 3 is listed (see Table 35, Note "c").

NOTE 2 For products eligible for gas testing, PSL 3G may be applied wherever PSL 3 is listed (see Table 35, Note “c”).

PSL Decision Tree

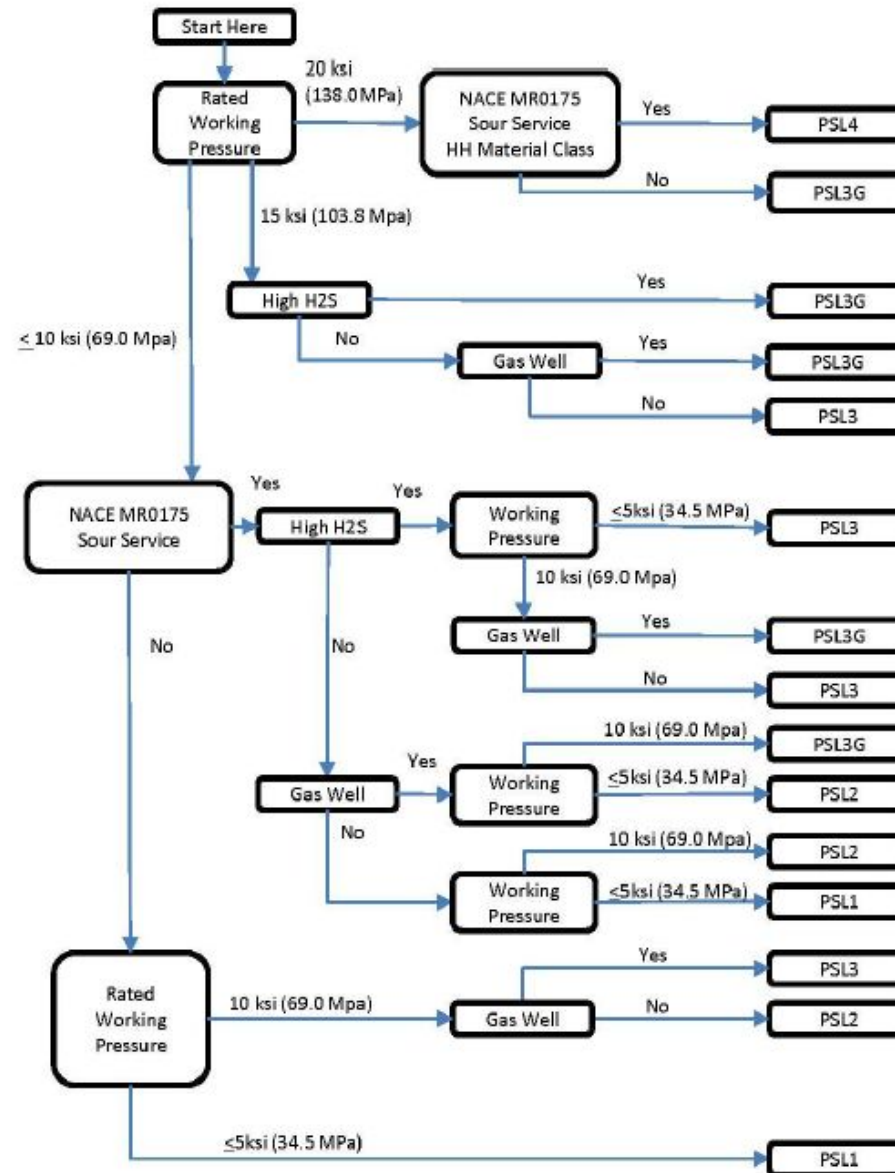


Figure B.15—Recommended Minimum PSL for Primary Parts



Performance Requirements

The performance requirements apply to all products being manufactured and delivered for service. The performance verification procedures are to be applied to designs of products, including design changes. Verification testing specified is intended to be performed on prototypes or production models. Per API 6A 21st Edition, there are three different performance requirement designations: PR1, PR2 and PR2F, with PR2F being applied to equipment passing the requirements of API 6A Annex F.

A design that undergoes a substantive change becomes a new design requiring performance verification. A substantive change is a change identified by the manufacturer which affects the performance of the product in the intended service condition. This may include changes in fit, form, function, or material.

PERFORMANCE REQUIREMENT (PR)



PR1:

This is the standard level of performance validation in API 6A. It ensures that the equipment meets the basic requirements based on material, temperature, and pressure specifications.

PR2:

This level represents a more rigorous validation process. It involves additional testing, such as pressure and temperature cycling, to demonstrate that the equipment can withstand the specified operating conditions.

PR2F:

PR2F is a PR2 level further validated by Annex F, requiring extensive testing and validation.



PR1 & PR2

Table 47—Operating Cycle Requirements for Valves

Performance Requirement Level	Operating Cycles
PR1	3 cycles
PR2	200 cycles

Table 50—Performance Requirements for Slip Hangers

Performance Requirement Level	Load Capacity
PR1	1 cycle at minimum ^a rated load to maximum rated load
PR2	3 cycles at minimum ^a rated load to maximum rated load
FOOTNOTE	
^a Minimum rated load may be upward or compressive.	



PR2F

Table F.6—Summary of Design Validation Requirements for Valves

Performance Requirement	PR2F
Open/close cycling dynamic pressure test at a temperature between 4 °C and 50 °C (between 40 °F and 120 °F)	160 cycles as specified in F.2.2
Low-pressure seat test at a temperature between 4 °C and 50 °C (between 40 °F and 120 °F)	1 h hold period at 5 % to 10 % of rated working pressure as specified in F.2.2
Open/close cycling dynamic pressure gas test at maximum and minimum temperatures	20 cycles at each extreme as specified in F.2.2
Low-pressure seat test at maximum and minimum temperatures	1 h hold period at 5 % to 10 % of rated working pressure as specified in F.2.2
Operating force or torque	As specified in F.2.2
Pressure/temperature cycling	As specified in F.1.11
Testing of nonmetallic seals	As specified in F.1.13



Wellhead Selection

Pressure Class: API holds user responsible for choosing class.

Temperature Class: API holds user responsible for choosing class.

Material Class: API holds user responsible for choosing class.

PSL (Product Specification Level): API recommends PSLs to the user.

PR (Performance Requirement): API requires cycle testing of designs

When ordering an equipment Purchaser has to use wellhead equipment data sheet.



Wellhead Selection

Selection Parameters

- Maximum Pressure
- Maximum Temperature
- H₂S
- CO₂
- Sand
- Production Method
- Completion Type

Engineering Checklist

- Casing Size
- Weight
- Grade
- Connection
- Load
- Burst
- Collapse
- Tension



Wellhead Selection

Wellhead Equipment Data Sheet—General

Well name(s) and location(s): _____

Maximum operating pressure: _____

Anticipated wellhead shut-in pressure: _____

Temperature ranges anticipated: _____

Minimum ambient temperature: _____

Maximum flowing fluid temperature at wellhead: _____

Anticipated composition of produced fluids: CO₂ _____ (mg) _____ Chlorides _____
(mg) _____ H₂ S _____ (mg) _____ Other _____

Water or brine pH: _____

Does NACE MR0175/ISO 15156 apply? _____

Anticipated production rates: _____ m³ /d oil/condensate
_____ m³ /d gas
_____ m³ /d S&W a

Will erosion be a concern? Cause: _____

Will scale, paraffin, corrosion, or other types of inhibitors be used? _____

Inhibitor type: _____ Inhibitor carrier: _____ Batch or continuous inhibition? _____

Will acidification be performed? _____ Type of acid: _____

External coating? Yes, type _____ No _____

Internal coating? Yes, type _____ No _____

Delivery requirements: _____

Special shipping, packing, and storage instructions: _____



Wellhead Selection

Wellhead Equipment Data Sheet—General

Casing program	Top Joint in String					
	Size (OD)	kg/m (lb/ft)	Grade	Connection	Max./Min. Casing Load daN (lb)	Bit Size
Conductor	_____	_____	_____	_____	_____	_____
Surface casing	_____	_____	_____	_____	_____	_____
Protective casing	_____	_____	_____	_____	_____	_____
Production casing	_____	_____	_____	_____	_____	_____
Tubing	_____	_____	_____	_____	_____	_____
Type of completion: single or multiple	_____	_____	_____	_____	_____	_____



Wellhead Selection

Common Mistakes

- Wrong Pressure Rating
- Wrong Material Class
- Wrong Bore
- Wrong Seal
- Improper Adapter
- Ignoring Thermal Effect

Failure Examples

- Seal Leakage
- Corrosion
- Improper Installation
- Collapsed Hanger
- Washout



CASE STUDY 1

- Land Well
- 20" Casing
- 13-3/8" Casing
- 9-5/8" Casing
- 7" Casing
- 2-7/8" Tubing
- Tubing Pressure 3,500 psi
- Question: Select WH and XT



CASE STUDY 2

- HPHT Well
- 12,500 psi
- 180°C
- Sour Gas
- Question
 - Material Class?
 - Temperature Class?
 - Pressure Rating?



SUMMARY

Summary

A good Wellhead Selection considers:

- Casing Design
- Well Pressure
- Temperature
- Production Fluid
- Corrosion
- Completion Method
- Applicable Standards
- Future Intervention Requirements

Q&A



Thank
You

